Highway Engineering

- INS3441 -

PROBLEM SESSION FOR ASSIGNMENT



Question

A state highway is designed as 2x1 lanes with 70 km/h design speed and 10 m platform width (2x(3.5+1.5)). At the beginning part of this highway two consecutive alignments intersect at a point S with a curve angle of $(84.04)^{o}$. Length of the first alignment is |AS|=300 m and a curve with radius R=150 m will be placed between the alignments.

Within this curve, superelevation design with ramp and transition curve should be performed by constant centerline principle. Clothoid will be used as transition curve. As a result of the traffic surveys there will be high amount of heavy vehicles in traffic (need for expansion). Perform superelevation design with all necessary mesurements for this curve.

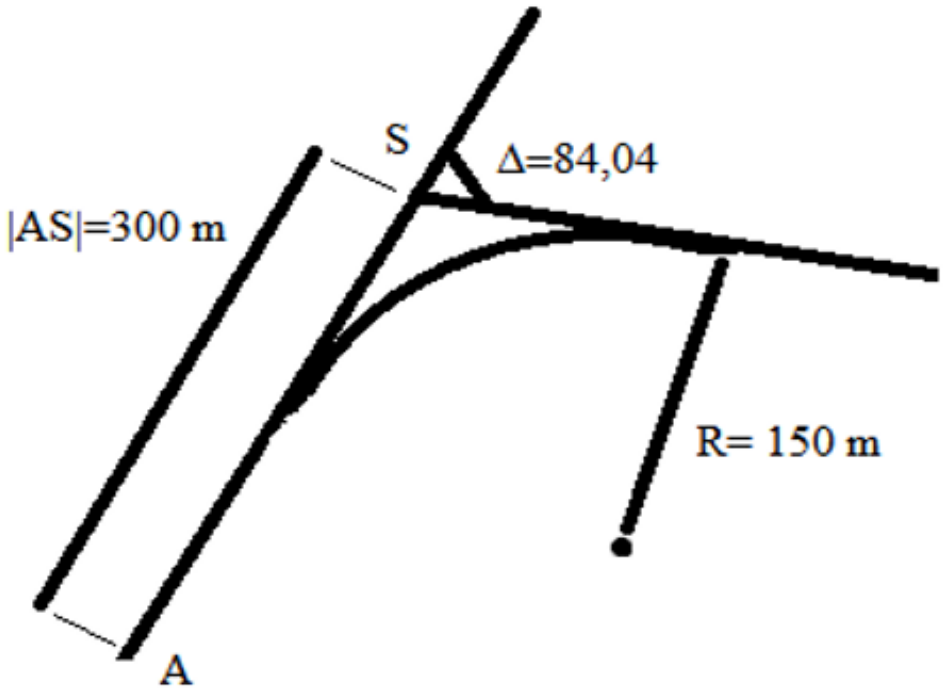
Given:

Crown slope = %2
Max. superelevation = %8
l *(vehicle length)* = 12 m

$q_{practical} = 0.443*(V_p^2 / R)$
Jerk (p‘) = $V^3 / (46.7*R*L) < 0.6$ m/sn³
Expansion value $(b_g) = (n*l^2 / 2*R) + (0.05*V_p / R)$
Ramp length $K=(2.h_1.L_g)/(h_2-h_1)$

Solution

Curve Geometry



$$D = \frac{2\pi R\Delta}{360} = \frac{2\pi * 150 * 84,04}{360} = 220,02m$$

$$t = R * tg(\frac{\Delta}{2}) = 150 * tg(\frac{84,04}{2}) = 135,16m$$

$$b = R * (\sec\frac{\Delta}{2} - 1) = 150 * (\sec\frac{84,04}{2} - 1) = 51,91m$$

$$\Phi_1(km) = \overline{AS} - t = 300 - 135,16 = 0 + 164^{84}$$

Superelevation

Max allowable superelevation is given as 8%

$$q_{pra} = 0,443 * \frac{V_p^2}{R} = 0,443 * \frac{70^2}{150} = 14,47 > 8$$

Therefore Select the superelevation as 8% and limit the speed.

New project speed:

$$8 = 0,443 * \frac{V_p^2}{150} \Rightarrow V_p = 52,05 \text{ km/h}$$

Selection of Clothoid Parameter

Optical condition:

$$\frac{R}{3} \leq A \leq R \Rightarrow \frac{150}{3} \leq A \leq 150$$

Dynamic condition:

$$A_{\min} = 0,19 * \sqrt{V_p^3} = 0,19 * \sqrt{52,05^3} = 71,35m$$

No need for superelevation condition.

Minimum length control:

L_g= at least 45 m.

$$L_g = \frac{A^2}{R} \Rightarrow L_{g,\min} = \frac{A_{\min}^2}{R} = 45$$

$$45 = \frac{A_{\min}^2}{150} \Rightarrow A_{\min} = 82,16m$$

So A is selected as A = 90 m. It is possible to select according to conditions above yet this value is selected in order to be safer.

$$L_g = \frac{A^2}{R} = \frac{90^2}{150} = 54m$$

Geometrical control :

$$\frac{L_g}{2} \le \frac{D}{2} \rightarrow \frac{54}{2} \le \frac{220,02}{2}$$

Jerk Control:

$$p' = \frac{V_p^3}{46,7 * R * L_g} = \frac{52,05^3}{46,7 * 150 * 54} = 0,37m/s^3 < 0,6m/s^3$$

Calculation of clothoid members

$$X = L - \frac{L^5}{40 * A^4} + \frac{L^9}{3456 * A^8} = 54 - \frac{54^5}{40 * 90^4} + \frac{54^9}{3456 * 90^8} = 53,83m$$

$$Y = \frac{L^3}{6 * A^2} - \frac{L^7}{336 * A^6} + \frac{L^{11}}{42340 * A^{10}} = \frac{54^3}{6 * 90^2} - \frac{54^7}{336 * 90^6} + \frac{54^{11}}{42340 * 90^{10}} = 3,23m$$

$$\tau = \frac{L}{2R} = \frac{54}{2 * 150} = 0,18rad \Rightarrow \tau = 0,18 * \frac{360}{2\pi} = 10,313°$$

$$X_m = X - R * \sin\tau = 53,83 - 150 * \sin(10,313) = 26,98m$$

$$Y_m = Y + R * \cos\tau = 3,23 - 150 * \cos(10,313) = 150,81m$$

$$\Delta R = Y_m - R = 150,81 - 150 = 0,81m$$

Calculation of clothoid members

$$T_k = \frac{Y}{\sin\tau} = \frac{3,23}{\sin(10,313)} = 18,04m$$

$$T_u = X - Y * \cot g(\tau) = X - Y * \frac{1}{\mathrm{t}\, g(\tau)} = 53,83 - 3,23 * \frac{1}{tg(10,313)} = 36,08m$$

$$\overline{TS} = X_m + (R + \Delta R) * tg\frac{\Delta}{2} = 26,98 + (150 + 0,81) * tg\frac{84,04}{2} = 162,86m$$

$$\Delta C = \Delta - 2 * \tau = 84,04 - 2 * 10,313 = 63,414°$$

$$DC = \frac{2\pi R\Delta C}{360} = \frac{2 * \pi * 150 * 63,414}{360} = 166,02m$$

$$\sigma = arctg(\frac{Y}{X}) = arctg(\frac{3,23}{53,83}) = 3,434°$$

$$S = \sqrt{X^2 + Y^2} = \sqrt{53,83^2 + 3,23^2} = 53,93m$$

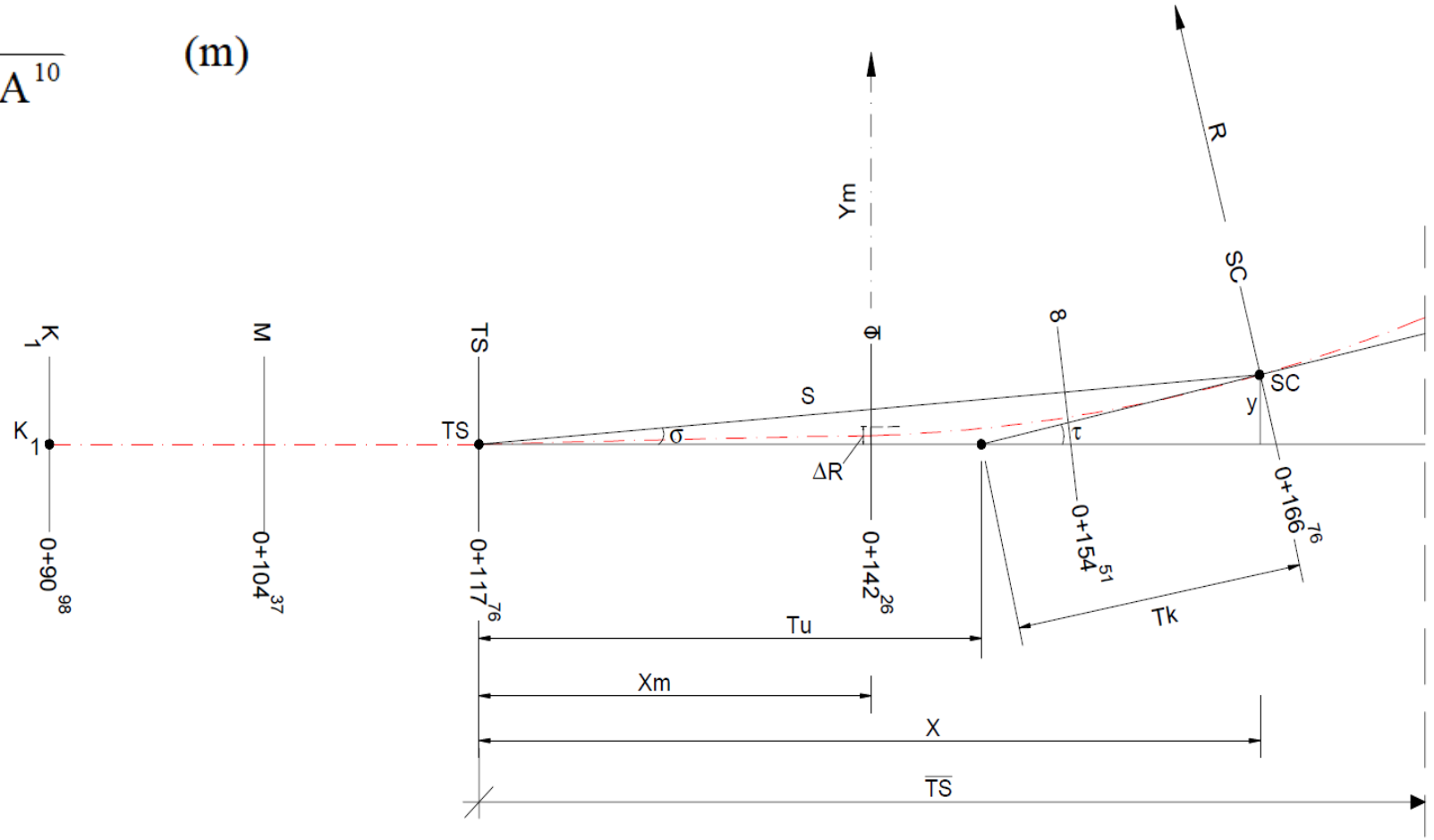
Calculate X and Y values based on

L=0, 5, 10, 15, 20, 25, 30, 35, 40, 45, …, L_g and the formulas:

$$X = L - \frac{L^5}{40 \cdot A^4} + \frac{L^9}{3456 \cdot A^8} \qquad (m)$$

$$Y = \frac{L^3}{6 \cdot A^2} - \frac{L^7}{336 \cdot A^6} + \frac{L^{11}}{42240 \cdot A^{10}} \qquad (m)$$

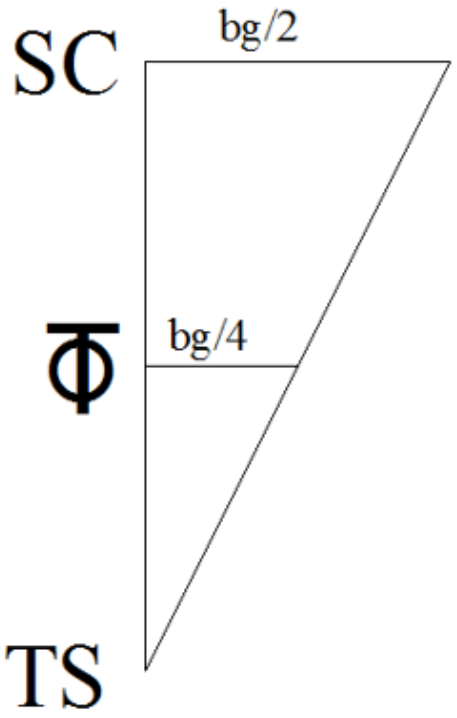
L (m)	X (m)	Y (m)
0	0	0
5	……	……
……	……	……



Expansion

R < 200 m so need for expansion. Expansion is equal on both sides of the road and it will increase linearly throughout the clothoid.

$$b_g = \frac{n*l^2}{2*R} + \frac{0,05*V_p}{\sqrt{R}} = \frac{2*12^2}{2*150} + \frac{0,05*52,05}{\sqrt{150}} = 1,17m$$



Both sides of the road will expand equally therefore it will start from zero and increase linearly until it reachs maximum expansion at SC point.

$$b_g(TS) = 0$$

$$b_g(\Phi) = \frac{b_g}{4} = \frac{1,17}{4} = 0,293m$$

$$b_g(SC) = \frac{b_g}{2} = \frac{1,17}{2} = 0,585m$$

Calculation of Ramp Length

$K = \frac{2 * L * h_1}{h_2 - h_1}$ formula can be used in order to calculate ramp length. h_1 and h_2 are required for this calculation. h_1 is the elevation difference at TS; no expansion here yet only roof slope. h_2 is the elevation difference at SC point where superelevation and expansion reach its maximum.

$$h_1 = q_1 * \frac{b}{2} = 0,02 * \frac{10}{2} = 0,1m$$

$$h_2 = q_2 * \frac{b + b_g}{2} = 0,02 * \frac{10 + 1,17}{2} = 0,447m$$

$$K = \frac{2 * 54 * 0,1}{0,447 - 0,1} = 31,12m$$

Superelevation Details

Calculation of Km values of critical points.

$$TS(km) = \Phi(km) - \frac{L}{2} = (0+164^{84}) - \frac{54}{2} = 0+137^{84}$$

$$K_1(km) = TS(km) - K = (0+137^{84}) - 31,12 = 0+106^{72}$$

$$SC(km) = TS(km) + L = (0+137^{84}) + 54 = 0+191^{84}$$

$$M_1(km) = TS(km) - \frac{K}{2} = (0+137^{84}) - \frac{31,12}{2} = 0+122^{38}$$

The other cross sections in that part will be showed. (for example; L/4 and 3L/4)

$$10(km) = TS(km) + \frac{L}{4} = (0+137^{84}) + \frac{54}{4} = 0+151^{34}$$

$$11(km) = \Phi(km) + \frac{L}{4} = (0+164^{84}) + \frac{54}{4} = 0+178^{34}$$

No need to show the other side of the curve on the superelevation diagram. Only calculations are necessary.

$$Ŧ(km) = Φ(km) + D = (0 + 164^{84}) + 220,02 = 0 + 384^{86}$$

$$CS(km) = Ŧ(km) - \frac{L}{2} = (0 + 384^{86}) - \frac{54}{2} = 0 + 357^{86}$$

$$ST(km) = Ŧ(km) + \frac{L}{2} = (0 + 164^{84}) + \frac{54}{2} = 0 + 411^{86}$$

$$K_2(km) = ST(km) + K = (0 + 411^{86}) + 31,12 = 0 + 442^{98}$$

$$M_2(km) = ST(km) + \frac{K}{2} = (0 + 411^{86}) + \frac{31,12}{2} = 0 + 427^{42}$$

Superelevation detail Drawing

All cross sections between K and SC points. Km values were defined before yet superelevation values for inner and outer edges should be calculated.

Expansion at critical points:

Expansion will reach its maximum value 1,17m at SC point. One side will expand 0,585 as equal to other side. This expansion will start from TS and increases linearly until SC. The length of the transition curve is 54 m therefore for each point expansion at each side will be 0,585 / 54 m.

$$b_g(TS) = 0$$

$$b_g(10) = [10(km) - TS(km)] * \frac{0,585}{54} = [(0+151^{34}) - (0+137^{84})] * \frac{0,585}{54} = 0,146m$$

$$b_g(\Phi) = [\Phi(km) - TS(km)] * \frac{0,585}{54} = [(0+164^{34}) - (0+137^{84})] * \frac{0,585}{54} = 0,293m$$

$$b_g(11) = [11(km) - TS(km)] * \frac{0,585}{54} = [(0+178^{34}) - (0+137^{84})] * \frac{0,585}{54} = 0,439m$$

$$b_g(SC) = 0,585m$$

Calculation of superelevations and elevation differences at critical points:

These calculations will be done based on linear increase of superelevation. There is only roof slope at point K, it will increase linearly until it reaches the max 8 % at SC point.

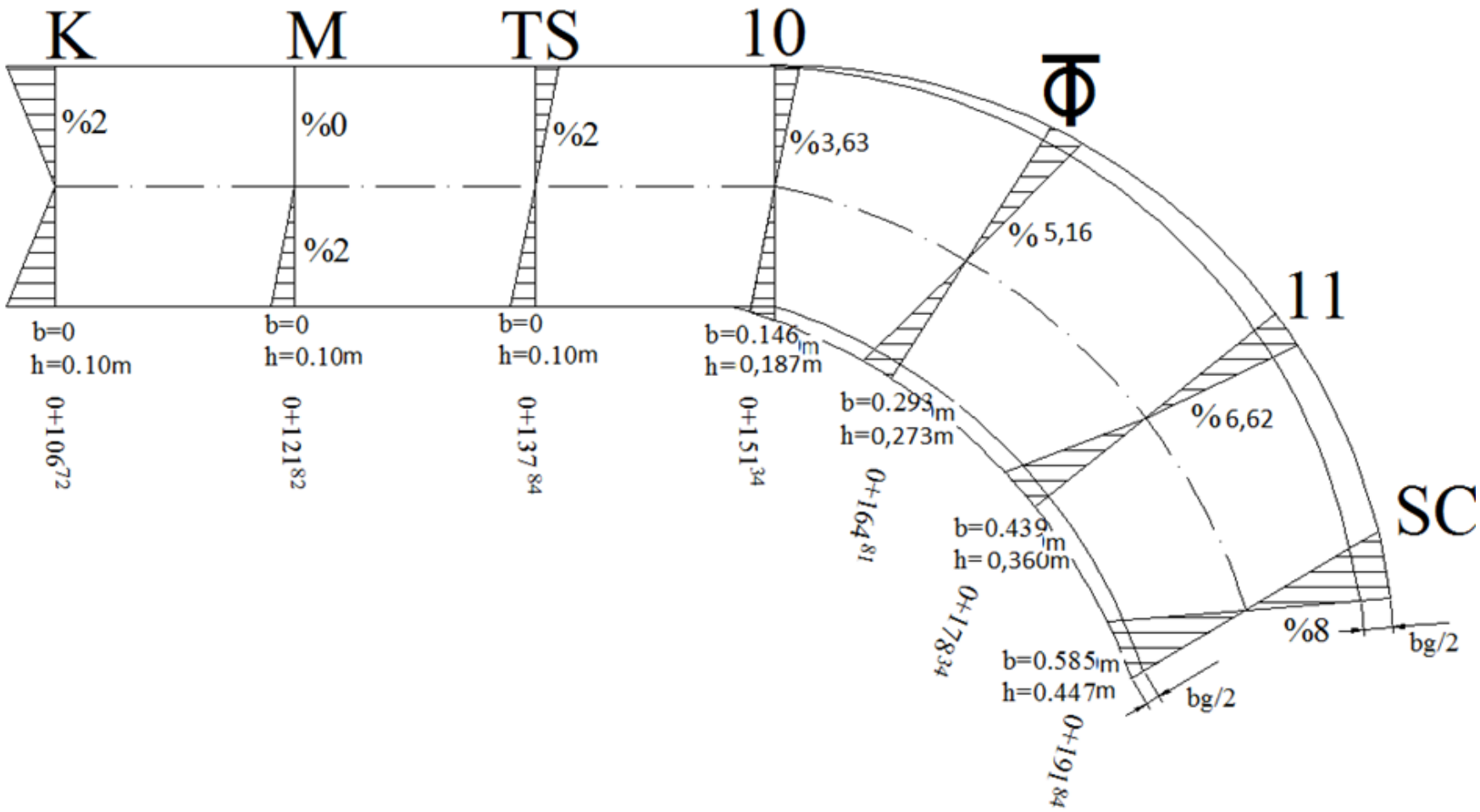
Elevation of outer edge will change from -0,1 m to +0,447 m within 85,12 m which is ramp + clotoid length.so for each meter elevation will increase 0,547 / 85,12 meters.

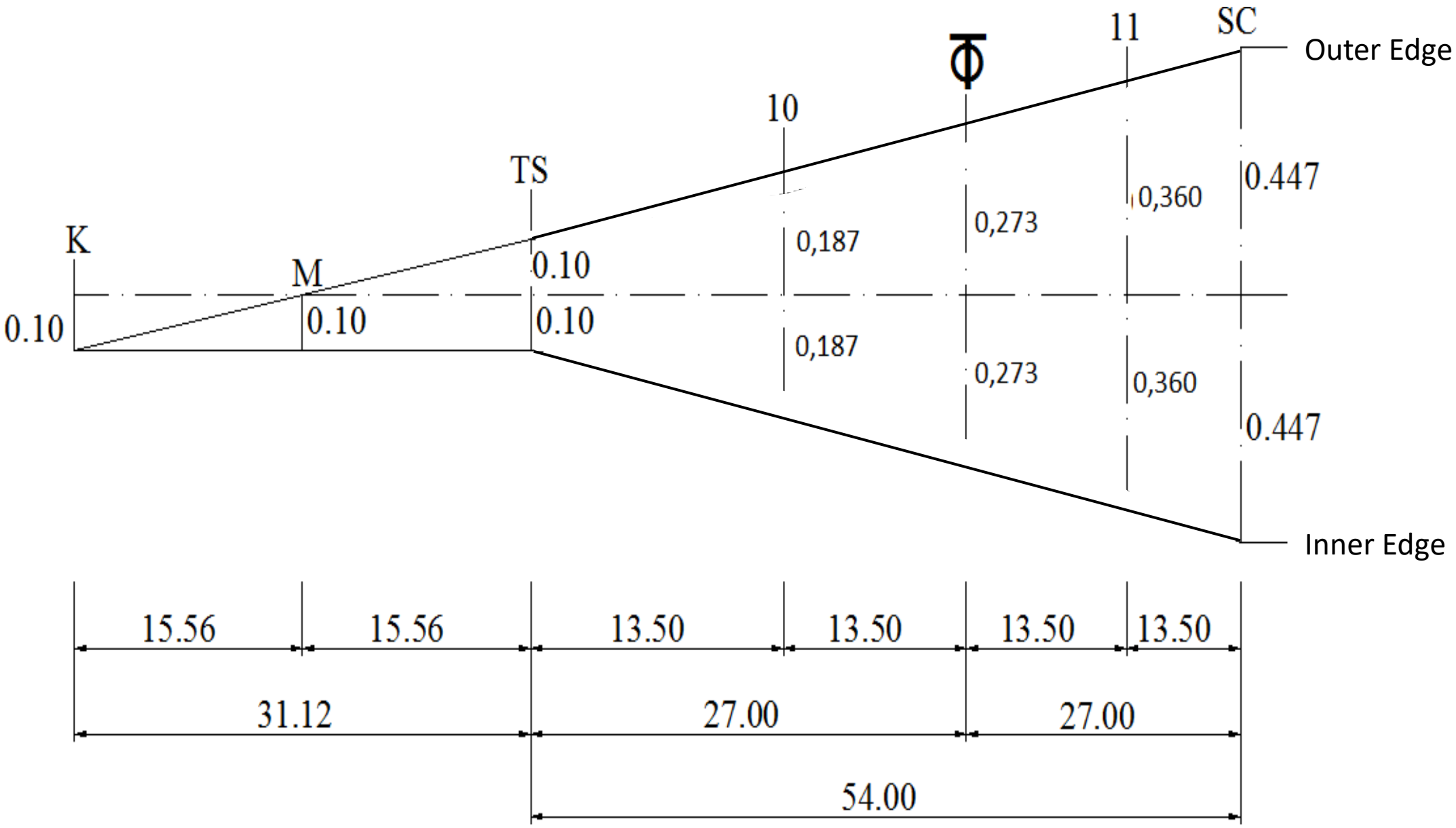
Superelevation and elevation calculations have been done based on this relation.

For example:

$$\frac{0{,}1 + 0{,}447}{85{,}12} = \frac{0{,}10 + h_{TO}}{58{,}12}$$

$$h_{TO} = 0{,}273m$$


K
M
TS
10
Φ
11
SC
%2
%0
%2
%2
%3,63
% 5,16
% 6,62
%8
bg/2
bg/2
b=0
h=0.10m
b=0
h=0.10m
b=0
h=0.10m
b=0.146m
h=0,187m
b=0.293m
h=0,273m
b=0.439m
h= 0,360m
b=0.585m
h=0.447m
0+106 72
0+121 82
0+137 84
0+151 34
0+164 81
0+178 34
0+191 84


K
M
TS
10
Φ
11
SC
Outer Edge
Inner Edge
0.10
0.10
0.10
0.10
0,187
0,273
0,360
0.447
0,187
0,273
0,360
0.447
15.56
15.56
13.50
13.50
13.50
13.50
31.12
27.00
27.00
54.00

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ASSIGNMENT

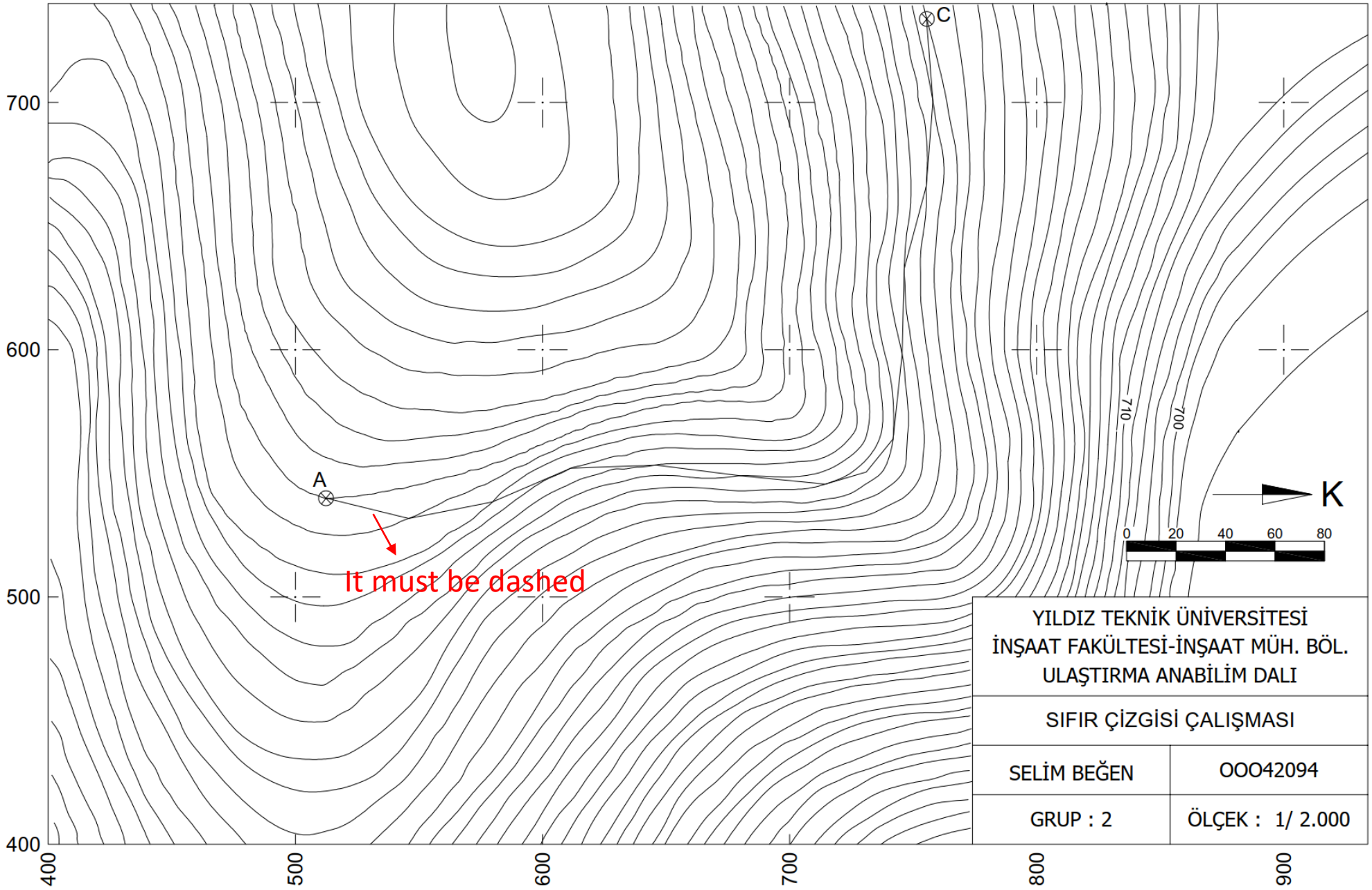


Final Submission

- Title page
- Calculations (*The route study, horizontal curve, transition curve, superelevation, the grade line on the profile, vertical curve, areas of cross-sections*)
- Zero line study sheet (Scale: 1/2.000)
- Route study sheet (Scale: 1/2.000)
- Precise axis calculation sheet (Scale: 1/2.000)
- Clothoid detail sheet (Scale: 1/500)
- Superelevation detail sheet (Scale; Horizontal: 1/500, Vertical: 1/10)
- Precise plan sheet (Scale: 1/2.000)
- Precise profile sheet (Scale; Horizontal: 1/2.000, Vertical: 1/200)
- Cross-section sheets (Scale: 1/100 or 1/200)

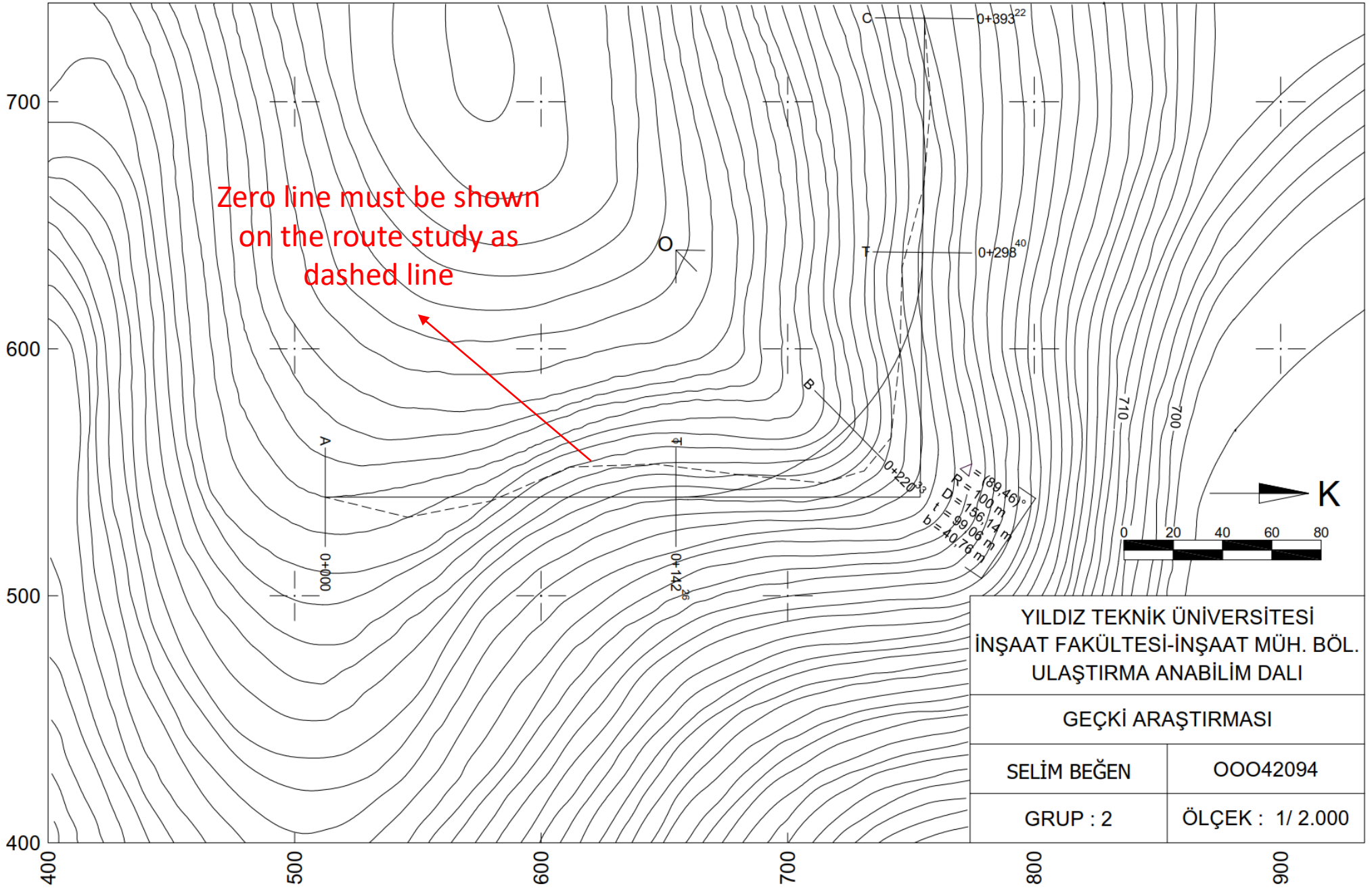
Zero Line Study Sheet

- Only zero line (as dashed line) will be shown on this sheet.


C
A
It must be dashed
700
600
500
400
400
500
600
700
800
900
710
700
K
0
20
40
60
80
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SIFIR ÇİZGİSİ ÇALIŞMASI
SELİM BEĞEN
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GRUP : 2
ÖLÇEK : 1/ 2.000

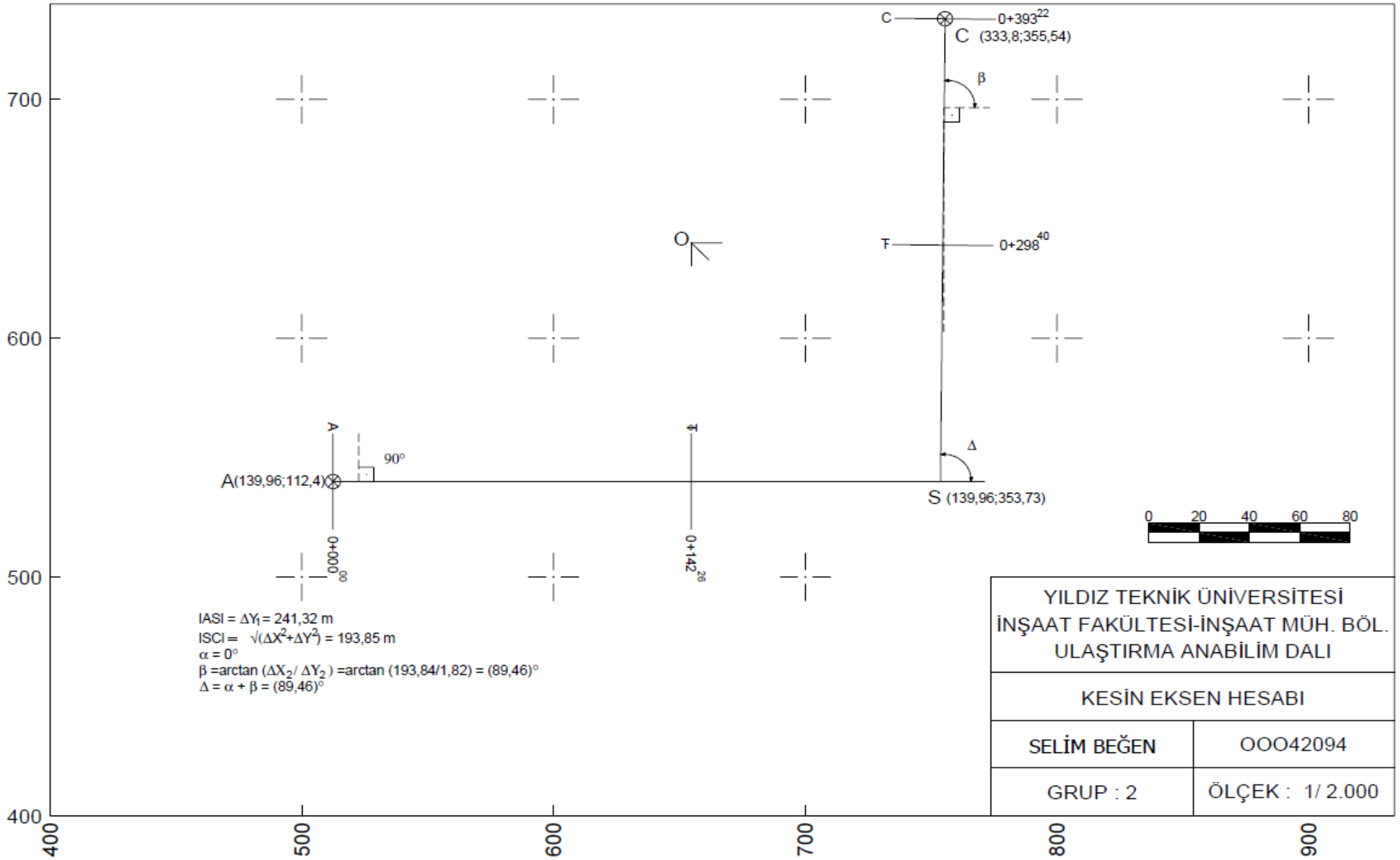
Route Study Sheet

- Cross-sections for route study:
  - The beginning (A) and ending (C) points of the highway
  - The beginning (Φ), ending (Ŧ) and the midpoint (B) of the horizontal curves
- Zero line must be shown on the route study as dashed line.
- The highway platform will not be drawn on this phase.


Zero line must be shown on the route study as dashed line
C
0+393,22
T
0+298,40
O
B
0+220,33
A
0+000
T
0+142,26
Δ = (89,46)°
R = 100 m
D = 156,14 m
t = 99,06 m
b = 40,76 m
710
700
K
0 20 40 60 80
700
600
500
400
400
500
600
700
800
900
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GRUP : 2
ÖLÇEK : 1/ 2.000

Precise Axis Calculation Sheet

- Precise axis calculation is calculation of the route length by using analytical geometry.
- See 3rd question in the 4th Problem Session.
- Y is horizontal axis, X is vertical axis.
- Prepare the precise axis calculation sheet without grid, contour lines and the arrow directed North.
- Also, the angles between that the alignments and the horizontal and vertical axis and curve angles should be shown symbolically.
- The sample calculations for angles and length of the alignments should be written on the map somewhere empty.


C
0+393^22
C (333,8;355,54)
β
O
T
0+298^40
A
90°
A(139,96;112,4)
Δ
S (139,96;353,73)
0+000^00
0+142^26
0 20 40 60 80
700
600
500
400
400 500 600 700 800 900
IASI = ΔY1 = 241,32 m
ISCI = √(ΔX²+ΔY²) = 193,85 m
α = 0°
β =arctan (ΔX2/ ΔY2) =arctan (193,84/1,82) = (89,46)°
Δ = α + β = (89,46)°
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KESİN EKSEN HESABI
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Clothoid and Superelevation Detail Sheet

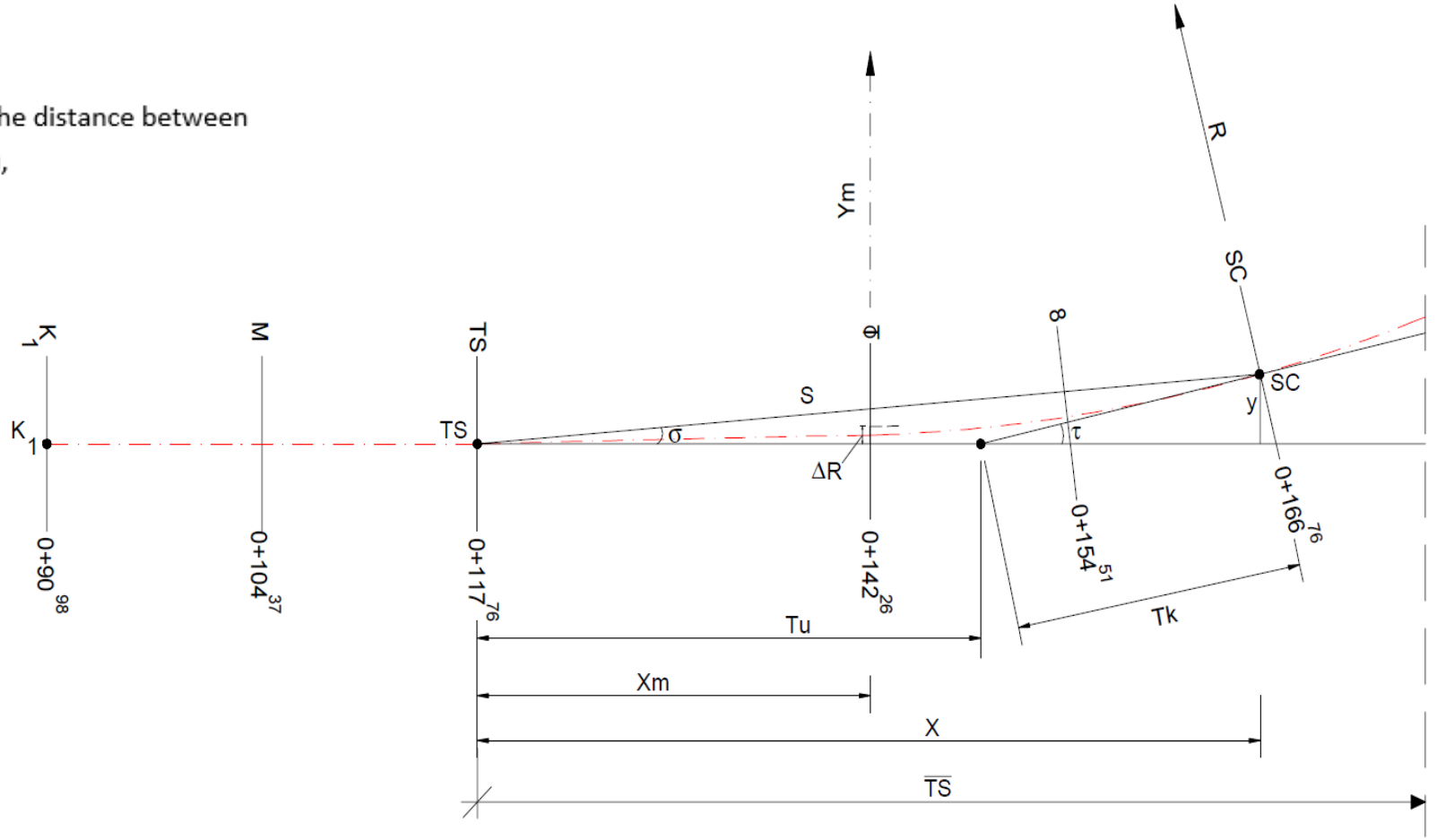
- Clothoid detail sheet (Scale: 1/500)
- Superelevation detail sheet (Scale; Horizontal: 1/500, Vertical: 1/10)

Calculation of Clothoid and Superelevation

- See problem session for assignment.

The elements of the clothoid can be listed as:

- L : Length of the clothoid (also can be notationed as L_t as the transition curve length) (m),
- X : The abscissa of the end point of the clothoid (SC) with respect to its starting point (TS) (m),
- Y : The ordinate of the end point of the clothoid (SC) with respect to its starting point (TS) (m),
- τ : The tangent angle of the clothoid (radians),
- X_m : The abscissa of the horizontal curve center (m),
- Y_m : The ordinate of the horizontal curve center (m),
- T_u : The length of the longer tangent of the clothoid (m),
- T_k : The length of the shorter tangent of the clothoid or the distance between the tangent of the curve to the common tangent point SC (m),
- ΔR : Displacement of the horizontal curve (m),
- S : Clothoid chord (m),
- σ : Direction angle of the clothoid chord (degree),
- $\overline{TS}$: The tangent length of the clothoid or the distance between the start and the PI of the horizontal curve (m),
- ΔC : Divergence angle of the shifted curve (degree),
- Dc : Length of the shifted curve (m) ($Dc=CS_{km}-SC_{km}$).



$$X = L - \frac{L^5}{40 \cdot A^4} + \frac{L^9}{3456 \cdot A^8} - \cdots \quad (m)$$

$$Y = \frac{L^3}{6 \cdot A^2} - \frac{L^7}{336 \cdot A^6} + \frac{L^{11}}{42240 \cdot A^{10}} - \cdots \quad (m)$$

$$\tau = \frac{L}{2 \cdot R} \quad \text{radian (convert to degree)}$$

$$X_m = X - R \cdot \sin\tau \quad (m)$$

$$Y_m = Y + R \cdot \cos\tau \quad (m)$$

$$T_u = X - Y \cdot \operatorname{cotg}\tau \quad (m)$$

$$T_k = \frac{Y}{\sin\tau} \quad (m)$$

$$\Delta R = Y_m - R \quad (m)$$

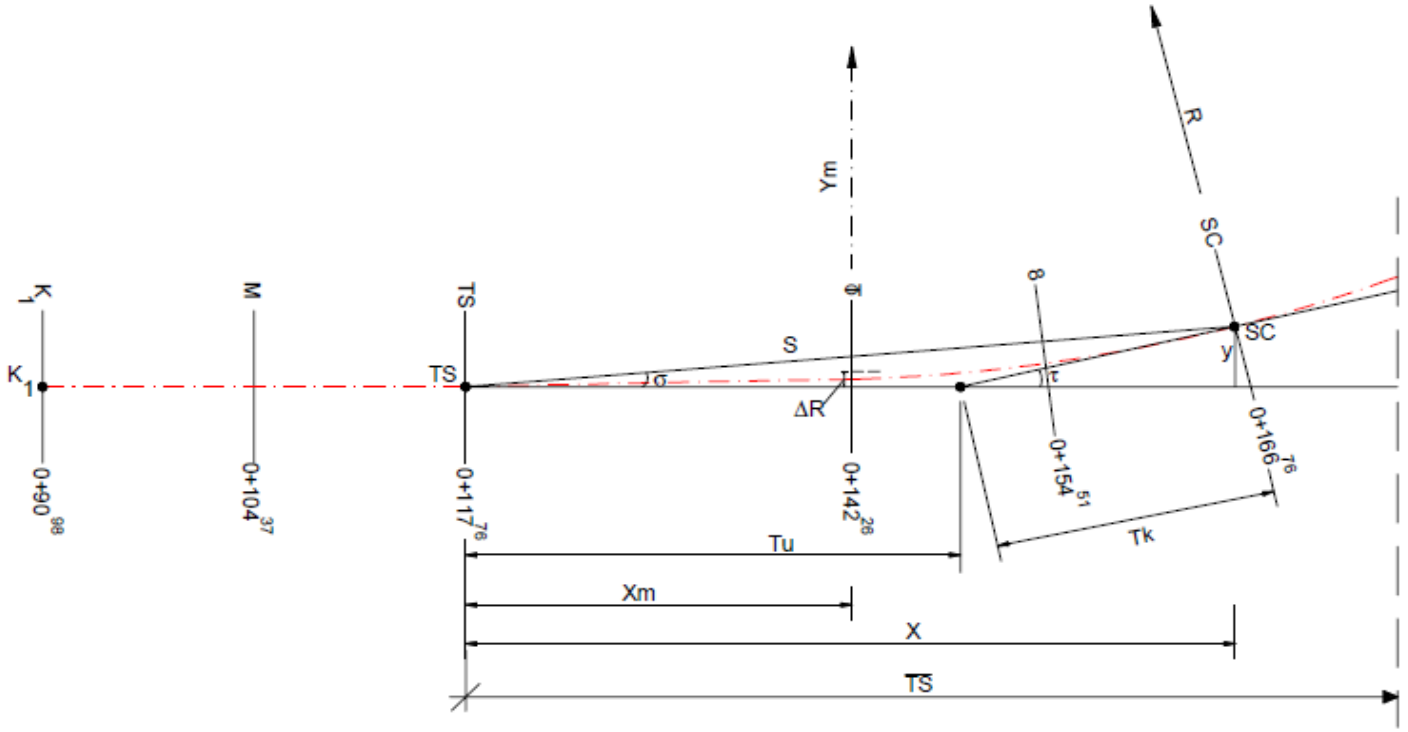
$$S = \sqrt{X^2 + Y^2} \quad (m)$$

$$\sigma = \operatorname{arctg}\left(\frac{Y}{X}\right) \quad (m)$$

$$\overline{TS} = X_m + (R + \Delta R) \cdot \operatorname{tg}\frac{\Delta}{2} \quad (m)$$

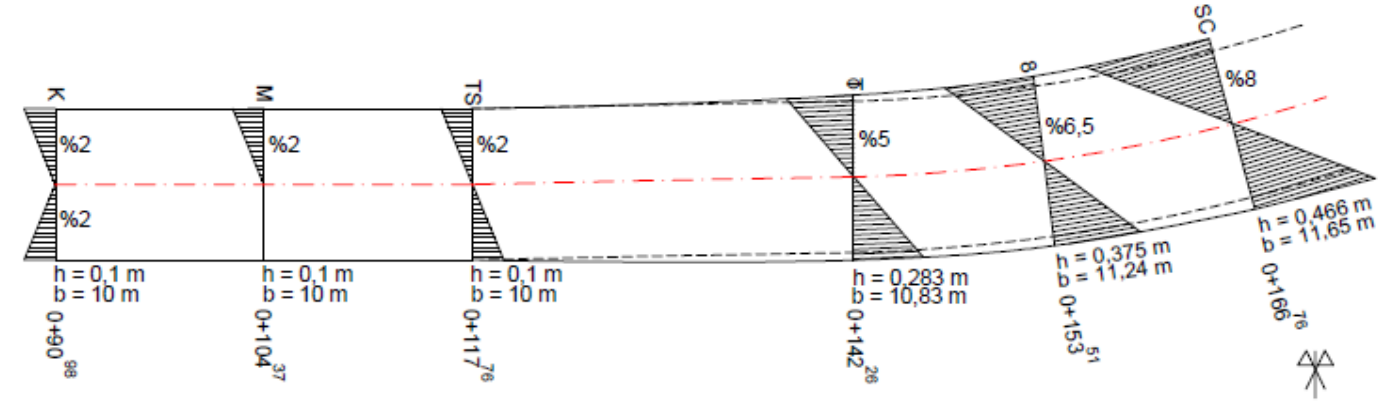
$$\Delta C = \Delta - 2 \cdot \tau \ \text{degree}$$

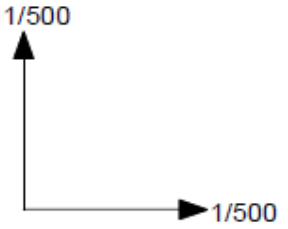
$$Dc = \frac{2 \cdot \pi \cdot R \cdot \Delta C}{360} \quad (m)$$

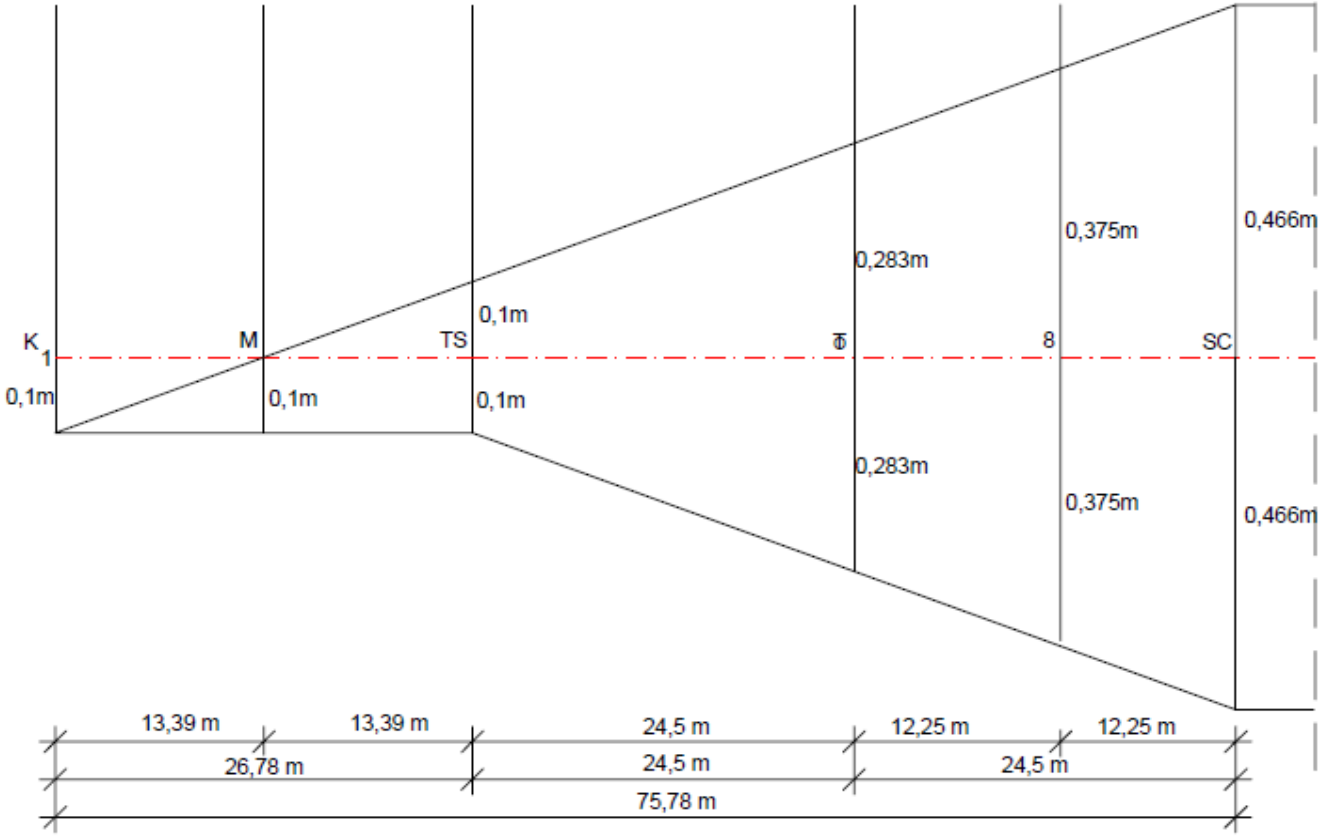


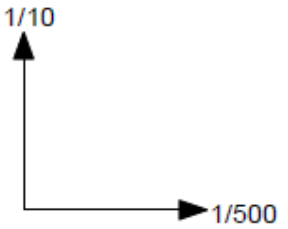
R = 100 m
L = 49 m
A = 70 m
X = 48,71 m
Y = 3,98 m
τ = 0,245 (radyan) = (14,04)°
Xm = 24,45 m
Ym = 100,993 m
Tu = 31,318 m
Tk = 16,406 m
ΔR = 0,993 m
S = 48,87 m
σ =(4,67)°
TS= 124,50 m
ΔC = (61,38)°
Dc = 107,13 m

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DEVER DETAYI	
SELİM BEĞEN	OOO42094
GRUP : 2	ÖLÇEK:1/10 - 1/ 500

Precise Plan Sheet

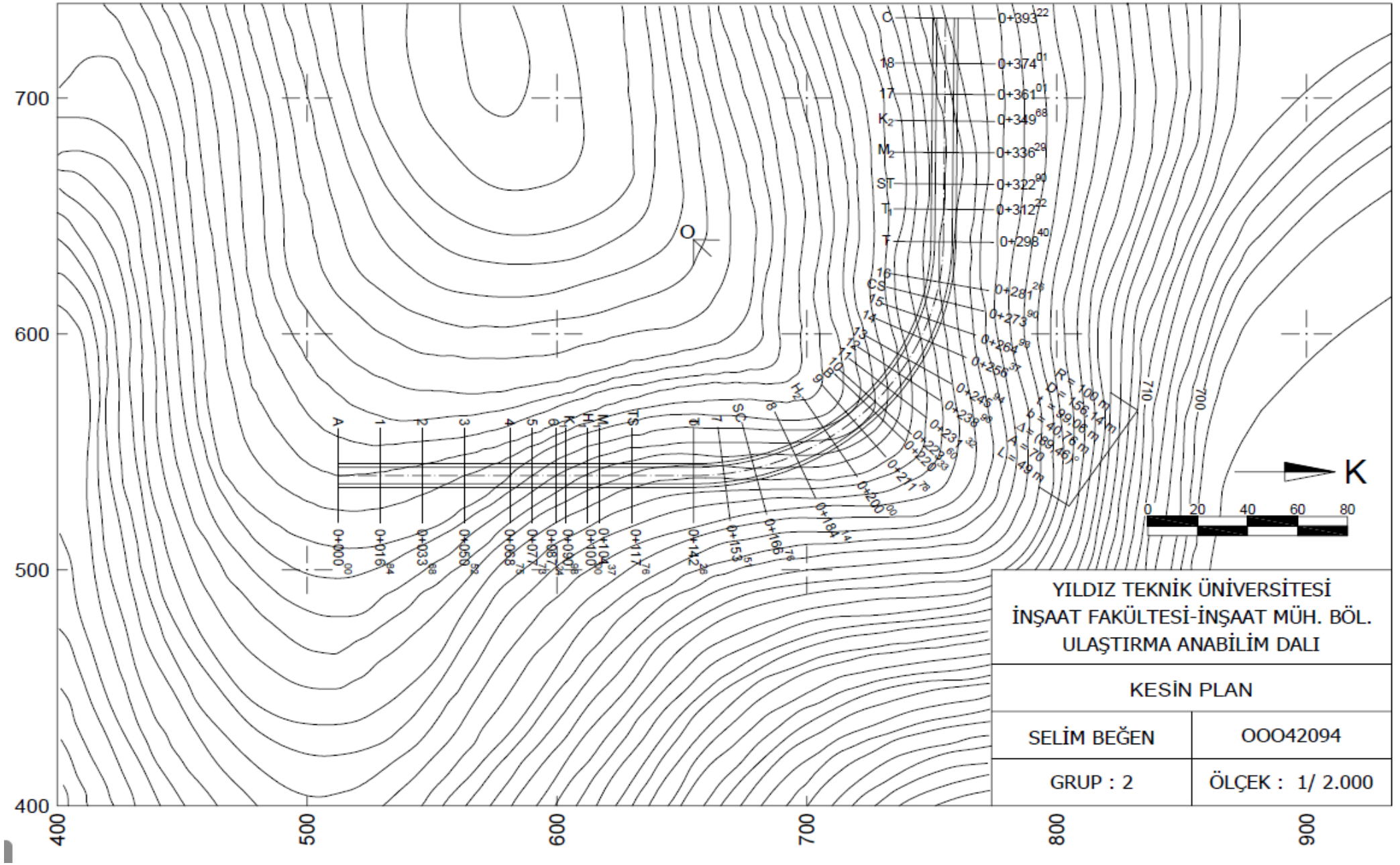
- The road width and the shoulders will be determined by drawing parallel lines on each side of the route axis. Road axis will be drawn as the dashed line while the shoulder and the platform borders will be drawn as continuous lines.

- On the precise plan, in addition to the cross-sections on the temporary plan, cross-sections will be drawn on TS, SC, CS, ST points of the transition curve; K_1, M_1, K_2, M_2 points of the ramp, T_1, P, T_2 points of the vertical curve.

- After all these cross-section lines are drawn, if some cross sections are intersecting with each other or are too close to each other (following each other with a distance less than 5 m), some of them will be cancelled according to their importance.

- The importance ➔ 1st group; starting and ending points of the highway and cross-sections of horizontal curves and cross-sections of vertical curves cross-sections of transition curves and cross-sections of ramp (if these are too close do not cancel them), 2nd group; hectometers and 3rd group; others.

- On horizontal curves, curve radius (R), angle of deflection (Δ), tangent length (t), curve length (L) and external distance (b) will be written on the outside the curve according to the direction of the highway, radially and one under the other. On horizontal curves which having a clothoid, clothoid parameter (A) and length (L_g) should be written as well.


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ULAŞTIRMA ANABİLİM DALI
KESİN PLAN
SELİM BEĞEN
OOO42094
GRUP : 2
ÖLÇEK : 1/ 2.000
K
0 20 40 60 80
400
500
600
700
800
900

Precise Profile

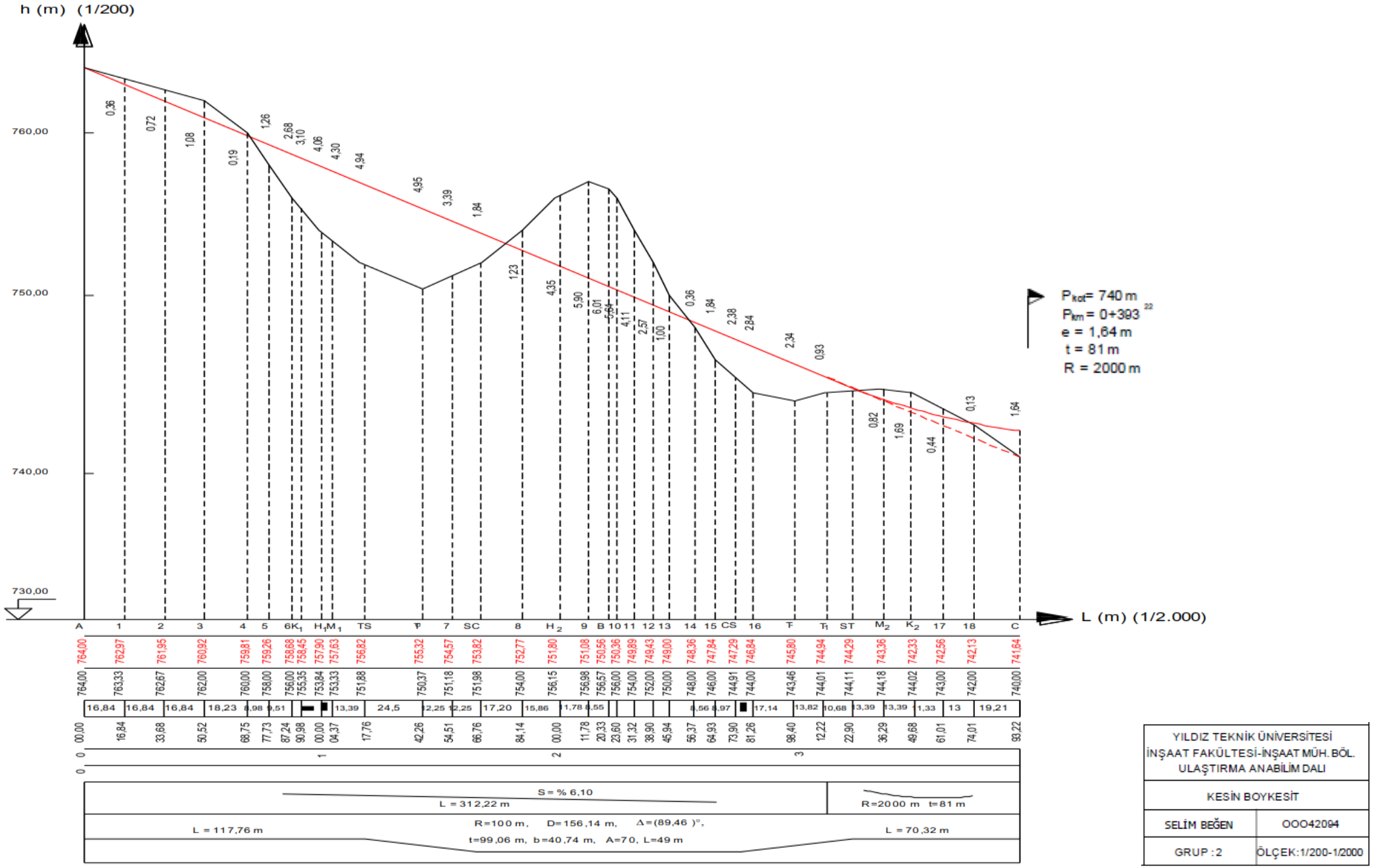
- One vertical curve must be added on the profile.
- Minimum radius for vertical curve; R_{CREST} = 4000 m, R_{SAG} = 2000 m
- Minimum vertical curve length; L= 120 m,
- The algebraic difference between the consecutive red line grades should fulfill the $G = g_1 - g_2 > \%0.5$ condition.
- A grade change sign (flag) will be placed on the places where the grade changes on the profile. At the right side of the flag, kilometer (P_{km}), elevation ($P_{elevation}$), bisector height (e), tangent length (t) and vertical curve radius (R) values will be written from top to bottom.

Precise Profile

- Even if the earthwork amount is too low or equalized to avoid the need for a vertical curve, a vertical curve will still be placed. In this case, such a way will be followed:

- If the red line has a DESCENDING grade, THE END of the highway will be connected to another red line having an ASCENDING grade. The grade of this red line will be taken as the last digit of the student number. In other words, if the last digit of the student number is n, the grade will be g_2=%n or g_2=0.0n. If the last digit of the student number is 0 (zero) n=1 will be taken to continue calculations.

- If the red line has a ASCENDING grade, THE END of the highway will be connected to another red line having an DESCENDING grade. The grade of this red line will be taken as the last digit of the student number. In other words, if the last digit of the student number is n, the grade will be g_2=%n or g_2=0.0n. If the last digit of the student number is 0 (zero) n=1 will be taken to continue calculations.



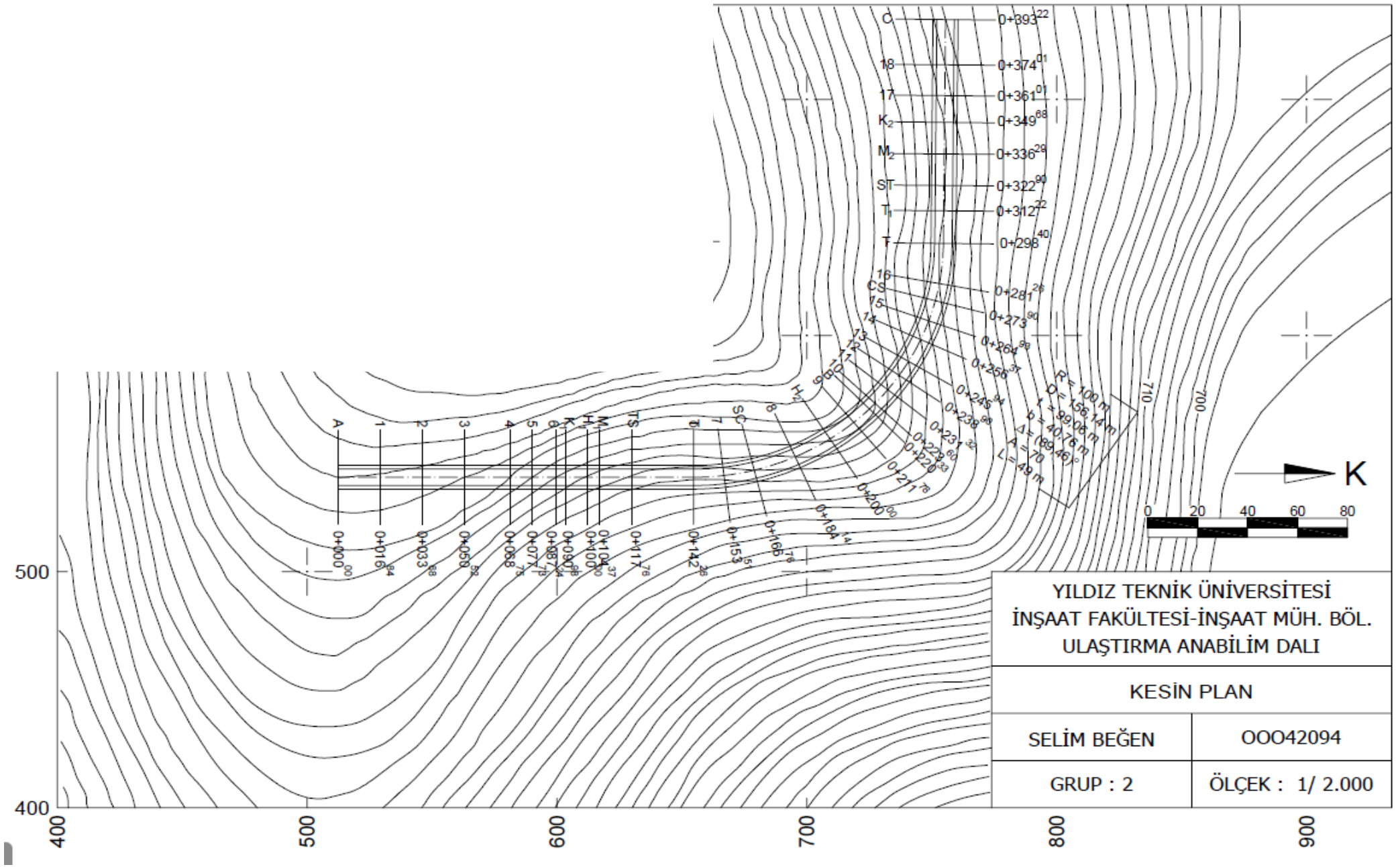
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KESİN BOYKESİT	
SELİM BEĞEN	OOO42094
GRUP : 2	ÖLÇEK:1/200-1/2000

Cross-sections

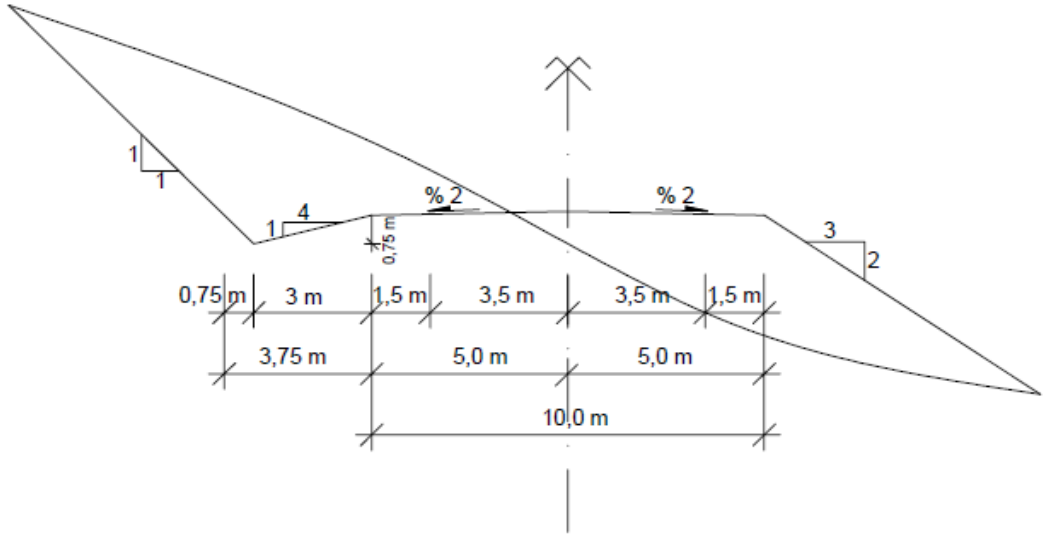
- Cross-sections of TS, SC, maximum cut point ve maximum fill point and typical cross-section (totally 5 cross-sections) will be drawn with a scale of 1/100 or 1/200.
- Write number, km value, red and black elevation values on each cross-section.
- Calculate the cross section areas by cross method and write them below the figure.
- If the cut and fill exceeds the allowable with draw retaining wall. You will find how to draw a retaining wall at the end of the lecture notes.

$$F = \frac{1}{2}\left[\sum y_i \cdot (x_{i-1} - x_{i+1})\right]$$



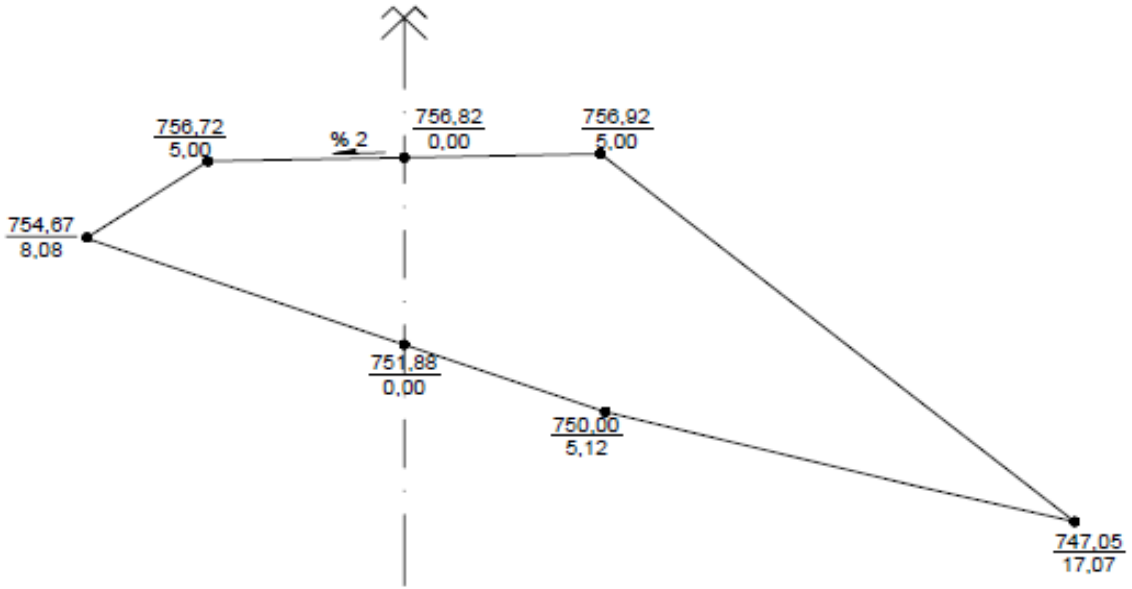

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KESİN PLAN
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OOO42094
GRUP : 2
ÖLÇEK : 1/ 2.000
K
0 20 40 60 80
400
500
600
700
800
900
710
700

TİP ENKESİT



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TİP ENKESİT	
SELİM BEĞEN	OOO42094
GRUP : 2	ÖLÇEK : 1/200

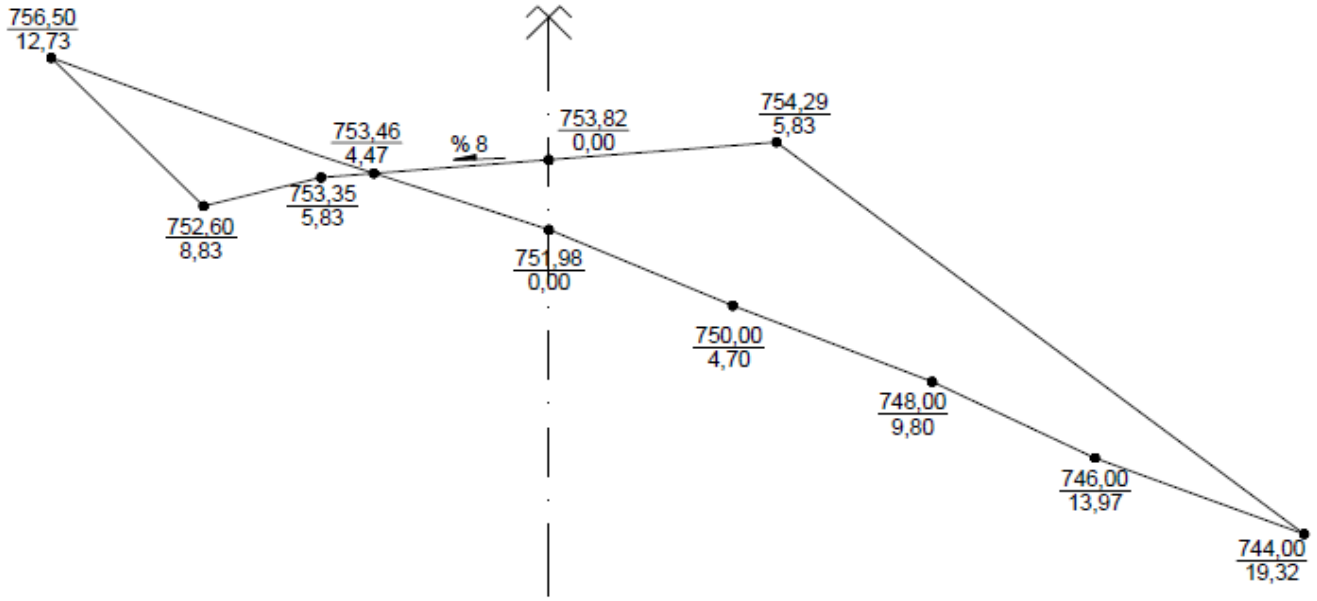
Enkesit No : TS
Km = 0+117[76]
S.kot = 751,88 m
K.kot = 756,82 m



Fd = 110,86 m²

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TS ENKESİT	
SELİM BEĞEN	OOO42094
GRUP : 2	ÖLÇEK : 1/200

Enkesit No : SC
Km = 0+166[76]
S.kot = 751,98 m
K.kot = 753,82 m



Fd = 65,37 m^2
Fy = 9,83 m^2

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SC ENKESİTİ	
SELİM BEĞEN	OOO42094
GRUP : 2	ÖLÇEK : 1/200

Important Notes

- The precise plan and precise profile must be submitted. If one of them is not submitted the assignment would not be evaluated (will take a score of 0)
- New cross-sections must be added and «Precise Plan» and «Precise Profile» must be written on the title block for Precise Plan and Precise Profile (and draw platform lines for precise plan).
- You must show all calculations on your assignment report. Drawings that do not have calculations in assignment report, will not be evaluated.